Experiment 1 — VHDL Code for 4-bit ALU

1. VHDL Design Code (alu4bit.vhd)

```
-- Module Name : alu4bit - Behavioral
-- Create Date : 07/30/2026
-- Design      : 4-bit Arithmetic Logic Unit (ALU)
----------------------------------------------------------------------

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_unsigned.ALL;

entity alu4bit is
    Port ( n1  : in  STD_LOGIC_VECTOR (3 downto 0);
           n2  : in  STD_LOGIC_VECTOR (3 downto 0);
           op  : in  STD_LOGIC_VECTOR (2 downto 0);
           res : out STD_LOGIC_VECTOR (3 downto 0);
           flag: out STD_LOGIC);
end alu4bit;

architecture Behavioral of alu4bit is
begin
    process(n1,n2,op)
    variable temp : std_logic_vector(4 downto 0);
    begin
        flag <= '0';
        case op is
            when "000" =>                       -- Addition
                temp := ('0' & n1) + n2;
                res  <= temp(3 downto 0);
                flag <= temp(4);                -- carry out
            when "001" =>                       -- Subtraction (|n1 - n2|)
                if (n1 > n2) then
                    res <= n1 - n2;
                else
                    res <= n2 - n1;
                end if;
            when "010" =>                       -- AND
                res <= n1 and n2;
            when "011" =>                       -- NAND
                res <= n1 nand n2;
            when "100" =>                       -- XOR
                res <= n1 xor n2;
            when "101" =>                       -- NOR
                res <= n1 nor n2;
            when "110" =>                       -- XNOR
                res <= n1 xnor n2;
            when others =>                      -- Pass n2
                res <= n2;
        end case;
    end process;
end Behavioral;
```

2. Constraints File (alu4bit.xdc / UCF)

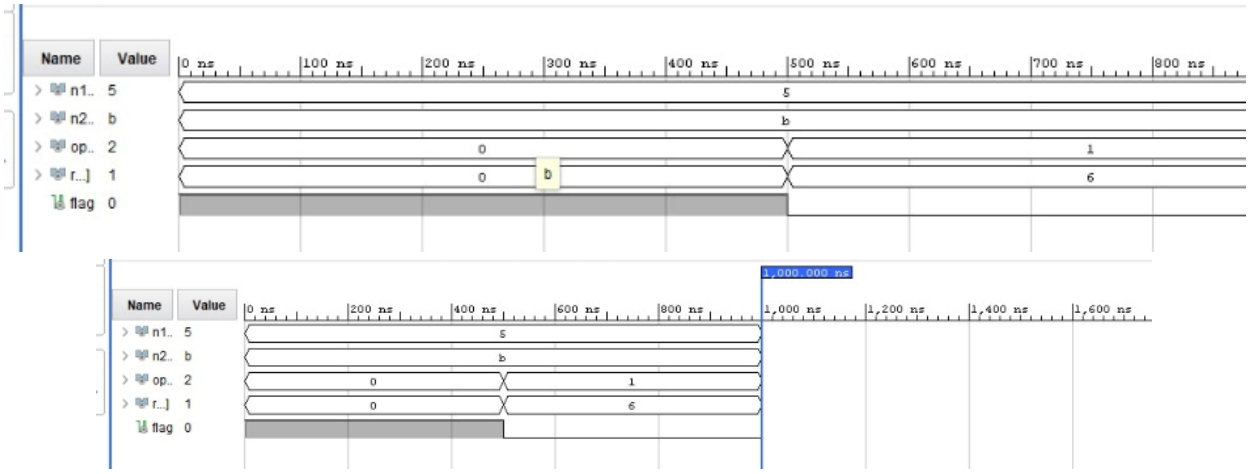
```
## Constraints file (alu4bit.xdc)  --  Basys 3 / Artix-7 (xc7a35tcpg236-1)

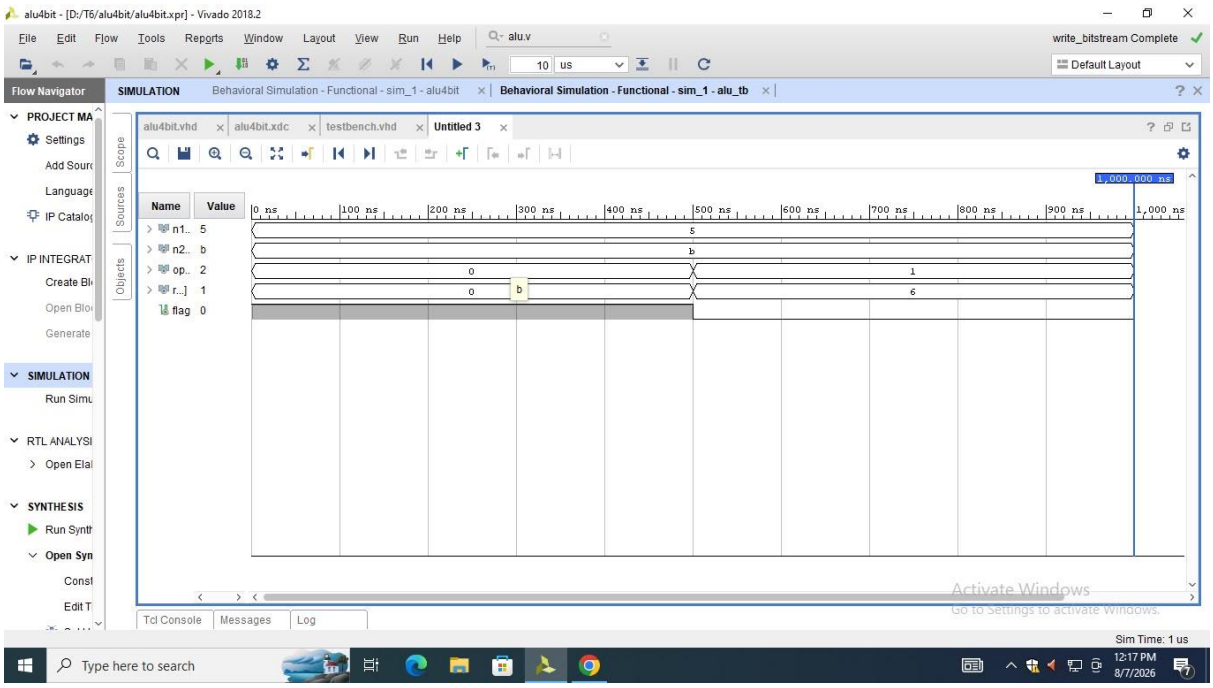
## I/O Standard
set_property IOSTANDARD LVCMOS33 [get_ports {n1[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {n1[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {n1[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {n1[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {op[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {op[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {op[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {n2[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {n2[2]}]
```

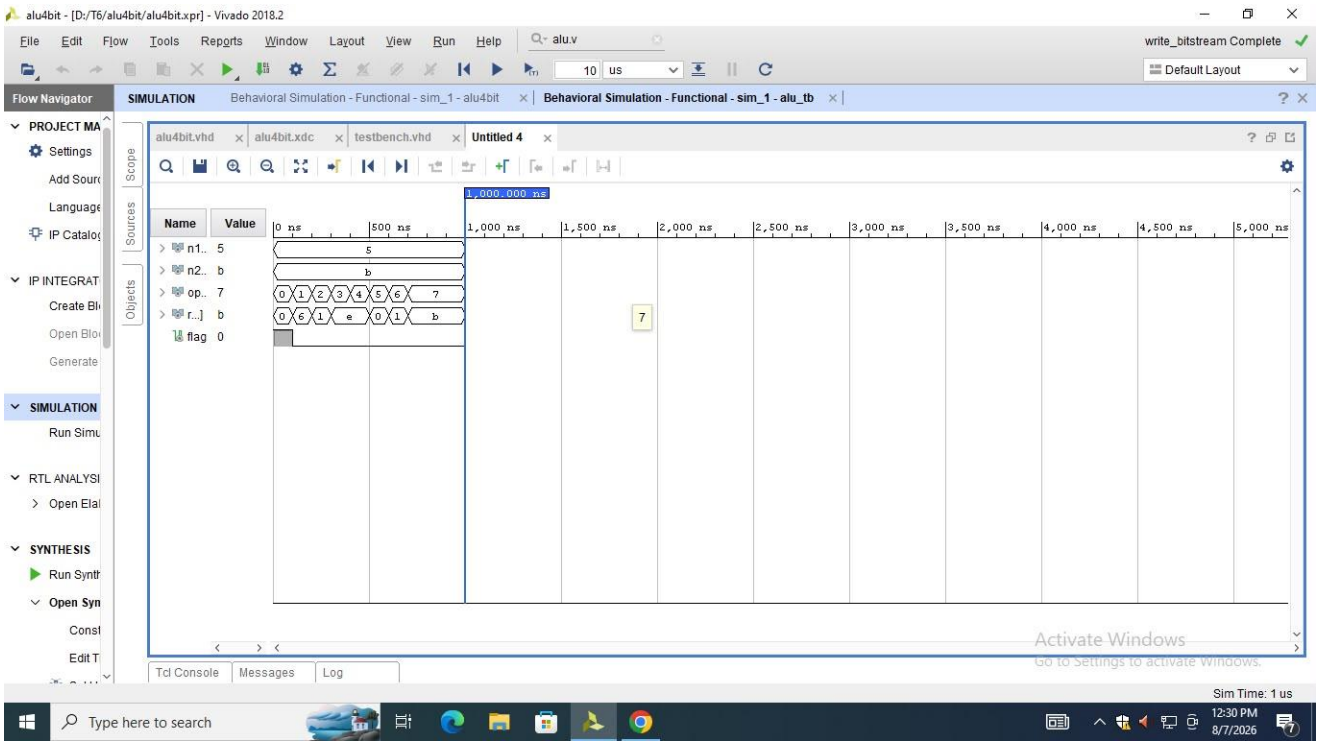
```
set_property IOSTANDARD LVCMOS33 [get_ports {n2[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {n2[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {res[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {res[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {res[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {res[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports flag]

## Package Pins
set_property PACKAGE_PIN R2  [get_ports {n1[3]}]
set_property PACKAGE_PIN T1  [get_ports {n1[2]}]
set_property PACKAGE_PIN U1  [get_ports {n1[1]}]
set_property PACKAGE_PIN W2  [get_ports {n1[0]}]
set_property PACKAGE_PIN R3  [get_ports {n2[3]}]
set_property PACKAGE_PIN T2  [get_ports {n2[2]}]
set_property PACKAGE_PIN T3  [get_ports {n2[1]}]
set_property PACKAGE_PIN V2  [get_ports {n2[0]}]
set_property PACKAGE_PIN W13 [get_ports {op[2]}]
set_property PACKAGE_PIN W14 [get_ports {op[1]}]
set_property PACKAGE_PIN V15 [get_ports {op[0]}]
set_property PACKAGE_PIN L1  [get_ports {res[3]}]
set_property PACKAGE_PIN P1  [get_ports {res[2]}]
set_property PACKAGE_PIN N3  [get_ports {res[1]}]
set_property PACKAGE_PIN P3  [get_ports {res[0]}]
set_property PACKAGE_PIN U3  [get_ports flag]
```

3. Simulation Waveform







4. VHDL Testbench Code (alu_tb.vhd)

```
--------------------------------------------------------------------------
-- Testbench : alu_tb.vhd   (for alu4bit)
--------------------------------------------------------------------------
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity alu_tb is
end alu_tb;

architecture Behavioral of alu_tb is

    component alu4bit
        Port ( n1  : in  STD_LOGIC_VECTOR (3 downto 0);
               n2  : in  STD_LOGIC_VECTOR (3 downto 0);
               op  : in  STD_LOGIC_VECTOR (2 downto 0);
               res : out STD_LOGIC_VECTOR (3 downto 0);
               flag: out STD_LOGIC);
    end component;

    signal n1   : STD_LOGIC_VECTOR (3 downto 0);
    signal n2   : STD_LOGIC_VECTOR (3 downto 0);
    signal op   : STD_LOGIC_VECTOR (2 downto 0);
    signal res  : STD_LOGIC_VECTOR (3 downto 0);
    signal flag : STD_LOGIC;

begin

    uut : alu4bit port map (
        n1   => n1,
        n2   => n2,
        op   => op,
        res  => res,
        flag => flag );

    stim_proc : process
    begin
        n1 <= "0101";      -- 5
        n2 <= "1011";      -- 11
        op <= "000";  wait for 500 ns;   -- ADD
        op <= "001";  wait for 500 ns;   -- SUB
        op <= "010";  wait for 500 ns;   -- AND
        op <= "011";  wait for 500 ns;   -- NAND
        op <= "100";  wait for 500 ns;   -- XOR
```

```
        op <= "101";  wait for 500 ns;   -- NOR
        op <= "110";  wait for 500 ns;   -- XNOR
        op <= "111";  wait for 500 ns;   -- pass n2
        wait;
    end process;

end Behavioral;
```

5. Synthesis Report (Selected Pages)

```
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----------------------------------------------------------------------------
| Tool Version : Vivado v.2018.2 (win64) Build 2258646
| Date         : Fri Aug  7 11:05:58 2026
| Host         : DESKTOP-22PHA3V running 64-bit
| Command      : report_utilization -file alu4bit_utilization_synth.rpt
| Design       : alu4bit
| Device       : 7a35tcpg236-1
| Design State : Synthesized
----------------------------------------------------------------------------

Utilization Design Information

1. Slice Logic
--------------
+-------------------------+------+-------+-----------+-------+
|        Site Type        | Used | Fixed | Available | Util% |
+-------------------------+------+-------+-----------+-------+
| Slice LUTs*             |   17 |     0 |     20800 |  0.08 |
|   LUT as Logic          |   17 |     0 |     20800 |  0.08 |
|   LUT as Memory         |    0 |     0 |      9600 |  0.00 |
| Slice Registers         |    0 |     0 |     41600 |  0.00 |
|   Register as Flip Flop |    0 |     0 |     41600 |  0.00 |
|   Register as Latch     |    0 |     0 |     41600 |  0.00 |
| F7 Muxes                |    0 |     0 |     16300 |  0.00 |
| F8 Muxes                |    0 |     0 |      8150 |  0.00 |
+-------------------------+------+-------+-----------+-------+
* Final LUT count, after physical optimization and full
  implementation, is typically lower.

4. IO and GT Specific
---------------------
+-----------------------------+------+-------+-----------+-------+
|          Site Type          | Used | Fixed | Available | Util% |
+-----------------------------+------+-------+-----------+-------+
| Bonded IOB                  |   16 |     0 |       106 | 15.09 |
| Bonded IPADs                |    0 |     0 |        10 |  0.00 |
| Bonded OPADs                |    0 |     0 |         4 |  0.00 |
| PHY_CONTROL                 |    0 |     0 |         5 |  0.00 |
| PHASER_REF                  |    0 |     0 |         5 |  0.00 |
| OUT_FIFO                    |    0 |     0 |        20 |  0.00 |
| IN_FIFO                     |    0 |     0 |        20 |  0.00 |
| IDELAYCTRL                  |    0 |     0 |         5 |  0.00 |
| IBUFDS                      |    0 |     0 |       104 |  0.00 |
| ILOGIC                      |    0 |     0 |       106 |  0.00 |
| OLOGIC                      |    0 |     0 |       106 |  0.00 |
+-----------------------------+------+-------+-----------+-------+

7. Primitives
-------------
+----------+------+---------------------+
| Ref Name | Used | Functional Category |
+----------+------+---------------------+
| LUT6     |   13 |                 LUT |
| IBUF     |   11 |                  IO |
| OBUF     |    5 |                  IO |
| LUT2     |    4 |                 LUT |
| LUT5     |    2 |                 LUT |
| LUT4     |    1 |                 LUT |
+----------+------+---------------------+

8. Black Boxes       : none
9. Instantiated Netlists : none
```

6. Implementation — I/O Planning & Device View

